

Instructor:



Reed Walker

Email: rwalker@berkeley.edu



Office Hours: Monday 4–5pm, Zoom (Meeting Link)

Graduate Student Instructor:



Silvia Farina

Email: sfarina@berkeley.edu



Office Hours: Tuesday 5-6pm Zoom (Meeting Link)

Course Objective

This course is designed to help you be an informed consumer and producer of data and quantitative analysis. Part of being an informed consumer is (1) understanding how and when to be skeptical, (2) understanding what makes analysis credible and convincing, and (3) understanding which questions you should be asking someone about their statistical work. Part of being an informed producer is (1) knowing which tools to use when, (2) knowing the limitations and pitfalls of certain tools, (3) being religious about data provenance / stewardship / cleanliness, and (4) being able to credibly communicate what you are doing and why. It is hard to generate expertise and mastery in all these areas in a single course, but we will try to build out some basic intuition for each.

At the same time, we don't want to bury ourselves in data and statistics without realizing what we hope to use them for. Each of our classes and topics will be closely connected to business settings and decisions through cases and examples.

Course Website

We will use bCourses for circulating powerpoint slides, cases, knowledge checks, and other materials. These materials are intellectual property of the Haas School of Business so please do

not repost. We will not be using *Study.Net* for this course.

Required Textbook

The required text is *Statistics for Business: Decision Making and Analysis, 3rd Edition*, by Robert Stine and Dean Foster, Pearson. The electronic version of the book will be provided.

Prerequisites

You are expected to understand the material covered in the pre-term. This material includes

1. Concepts of a mean (average), median, variance, and standard deviation.
2. Different data and variable structures: categorical variables, time series data, cross-sectional data
3. Distributions of data: what they look like and how to discuss attributes like skewness
4. Basic ways of summarizing data and relationships: scatterplots, correlation, covariance
5. Understanding the properties of the normal distribution and how to calculate probabilities from the normal distribution

You can review these concepts in Chapters 1-11 of the Stine and Foster textbook, use material from pre-term, and/or lots of material on the internet.

Team Exercises

The course will have several, hands-on data exercises that will be done in teams. Some of these exercises will be done in class while other exercises will be assigned between blocks. All of these will be turned in and graded and will count towards your final grade in the course. The program office will assign teams prior to Block 1.

Software

Team data exercises are all possible in Microsoft Excel but will require the “Analysis ToolPak”. To activate the Toolpak, from Excel Options go to “Add-Ins” to activate. If you are using a Mac, please be sure to use a current version of Microsoft 365 for Mac (available for free to all Haas students), as the ToolPak is not available in every Mac version. Students may also use other statistical software, such as R, Stata, or Python if interested. ChatGPT, Claude, or other AI tools are great resources for learning code/implementation of these programming languages for those looking to learn. Moreover, modern AI tools, when paired with statistical software (e.g., Claude Code, Codex, or Cursor connected to an R or Python installation on your own machine), function less like a chatbot and more like a research assistant that can run the analysis end to end. I’d encourage everyone to explore these latter capabilities prior to class, as these tools can significantly accelerate learning/implementation. We have also posted basic tutorials to using these software on bCourses.

AI Policy

Generative AI tools (ChatGPT, Claude, Gemini, Claude Code, Codex, Cursor, Copilot, etc.) are permitted and encouraged on all team exercises, both in-class and between blocks. Part of the point of this course is to learn how to use these tools well, so you should treat them as you would any other resource: use them to write code, explore data, check your reasoning, and draft explanations. You remain fully responsible for the accuracy and integrity of what you submit. “The AI told me so” is not a defense for a wrong answer, and you should be able to explain and defend every number and claim in your work.

Acknowledging AI use Each team submission should include a brief AI-use statement (a few sentences is plenty) describing which tools you used and what you used them for—e.g., “We used Claude Code to clean the data and run the regressions in Python, and ChatGPT to check our interpretation of the confidence interval.” You do not need to include prompts or transcripts. If you did not use AI on an assignment, say so. The statement is not graded and does not affect your score; its purpose is to help us understand how the tools are being used so we can teach to it.

Final exam The final exam is a paper exam taken without laptops, phones, or other devices, so AI tools are not available and may not be used. See the Final Exam section below.

Knowledge Checks

Prior to each block (including Block 1), we will post an online knowledge check that will be available via bCourses. You should feel free to use this as a self-assessment that you are staying current with the material and understanding the basic concepts / applications. These will not be for a grade, but you will receive a score and/or an answer sheet to help with the self-assessment.

Office Hours and Discussion Sections

The Graduate Student Instructor (GSI) will deliver a 1.5 hour discussion/review section between blocks in addition to weekly office hours. The main purpose of these discussion sections is to review material from the previous block. If there are particular topics you want the GSI to cover, please let them know well in advance of the session so they may prepare. These sections will be recorded and posted to bCourses.

I will be holding weekly office hours via Zoom on Mondays from 4–5pm (link above). If that time does not work for you, email me and we will find another. Thanks to our class representatives, BK Abebe and Sushruth Gade, for coordinating on scheduling.

Midterm Exam

There is **no** midterm.

Final Exam

The final exam is scheduled for Friday, November 13th, 1:00–4:00pm (Session 10). The final exam will be a paper exam, open book and open note, completed within the scheduled class

time. Laptops, phones, and other electronic devices are not permitted; bring a calculator and any printed notes you wish to use. The final exam is an *individual* assignment so there will be no communication between class participants during the final exam for any reason.

Grades

Performance will be graded as follows:

<u>Activity</u>	<u>Percent of Final Grade</u>
In-class Team Assignments	10%
Between Block Team Assignments	40%
Final Exam	40%
Attendance and Participation	10%

Lecture Recordings

In-class sessions will be recorded. See this Privacy and Recording Notice for more details.

Grade Dispute Policy

In the event that a team or individual requests that an assignment or exam be re-graded, within 72 hours of it being returned, class members can submit a petition in writing. The petition should clearly document the specific issue you have with the grading, be well reasoned, and be word-processed. Please turn in the written petition and the assignment/exam at the same time. Please do not write on the assignment/exam, but rather on a separate piece of paper. We reserve the right to re-grade the whole assignment/exam.

Honor Code

As members of the UC-Berkeley community, our expectation is that you will adhere rigorously to the UC-Berkeley Honor Code. Anyone caught cheating on an assignment or exam in this course will receive a failing grade in the course and will also be reported to the University Center for Student Conduct. For further information see: www.asuc.org/honorcode/.

Use of AI tools is governed by the AI Policy above. Using AI (or any electronic device) during the final exam is a violation of the honor code and will be treated as cheating.

Accommodations

If you need disability-related accommodations in this class, if you have emergency medical information you wish to share with me, or if you need special arrangements, please let me know during the first week of classes.

XMBA200S: Data and Decisions, Class Schedule

#	Date	Time	Room	Topic	Chapters
<i>Block 1</i>					
1	Thu 9/10	1:00–4:00pm	N100	Course Introduction; Samples and Surveys	13
2	Fri 9/11	1:00–4:00pm	N300	Sampling Variation and Quality	14.1, 14.2
3	Sat 9/12	8:30–11:30am	N300	Confidence Intervals	15
<i>Block 2</i>					
4	Thu 10/1	8:30–11:30am	N100	Statistical Tests	16
5	Fri 10/2	1:00–4:00pm	N300	Comparison and A/B Testing	17
6	Sat 10/3	8:30–11:30am	N100	Linear Patterns and Curved Patterns	19, 20.1, 20.4
<i>Block 3</i>					
7	Thu 10/22	8:30–11:30am	N300	The Simple Regression Model and Multiple Regression	21, 23
8	Fri 10/23	8:30–11:30am	N300	Building Regression Models and Machine Learning	24
<i>Block 4</i>					
9	Thu 11/12	1:00–4:00pm	N300	Categorical Variables, Experimental Design, Wrap-up	25
10	Fri 11/13	1:00–4:00pm	N300	Final Exam	—